

A Strict Separation Between Spectral-Tail and Orbit-Localization Clocks

Bin Wang

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Abstract

The missing weighted bridge might appear approachable by extending the fixed-length Gaussian localization theorem to a logarithmic moving order. We show that the current architecture cannot overlap the projection-free tail clock. The Hilbert–Schmidt mass-and-cap tail at $q = 1/2$ and $R = 7/5$ requires slope at least $1/\log(10/7) = 2.803673\dots$. Boundary crowding forces the existing interior orbit-tube proof to have slope at most $1/\log \lambda = 1.930709\dots$. The two closed slope ranges are strictly disjoint. This is a rigorous proof-method obstruction. It does not rule out actual trace convergence obtained by a new moving boundary-layer or aggregate determinant argument.

1 The two clocks

Write $L_\sigma = \log(1/\sigma)$ and $m_\sigma = \lceil aL_\sigma \rceil$. The RH-283 mass-and-cap theorem gives the critical tail slope

$$a_{\text{tail}} = \frac{1}{\log(1/(qR))} = \frac{1}{\log(10/7)} = 2.803673252057129\dots$$

At equality its certified mass-and-cap upper bound is $O(L_\sigma^{-1})$.

For a proposed prefix cut m_σ , let

$$\ell_\sigma = 2 \left\lfloor \frac{m_\sigma - 1}{2} \right\rfloor$$

be the largest even order below it. The boundary-cycle theorem gives cycles at this even order with clearance

$$\Delta_{\ell_\sigma} \leq C\lambda^{-\ell_\sigma}, \quad \lambda = 1.678573510428322\dots$$

A proof uniform over every order below m_σ must handle this even order. The existing Gaussian orbit-tube argument requires at least $\Delta_{\ell_\sigma} \gtrsim \sigma$, and its asymptotic form requires $\Delta_{\ell_\sigma}/\sigma \rightarrow \infty$. Thus its limiting prefix-slope ceiling is

$$a_{\text{loc}} = \frac{1}{\log \lambda} = 1.930709419186936\dots$$

2 Separation theorem

Theorem 2.1. *There is no logarithmic slope a that is both tail-admissible for the uniform RH-282 mass-and-cap estimate and admissible for the archived interior orbit-tube localization proof. Quantitatively,*

$$a_{\text{tail}} - a_{\text{loc}} = 0.872963832870193\dots > 0, \quad \frac{a_{\text{tail}}}{a_{\text{loc}}} = 1.452146669092140\dots$$

At the maximal localization slope the tail power exponent is

$$a_{\text{loc}} \log(10/7) - 1 = -0.3113643261 \dots,$$

so the mass-and-cap upper bound does not vanish. At every $a \geq a_{\text{tail}}$,

$$\frac{\Delta_{\ell_\sigma}}{\sigma} \leq C' \sigma^{a \log \lambda - 1} \longrightarrow 0.$$

Proof. The first two displays are direct substitutions. Since $a_{\text{tail}} > a_{\text{loc}}$, the slope intervals do not meet. Also $a_{\text{tail}} \log \lambda = 1.452146 \dots > 1$, so the clearance ratio tends to zero at the minimal tail slope and therefore at every larger slope; here $\ell_\sigma = aL_\sigma + O(1)$ absorbs parity and rounding into C' . $\square$

3 Protocol and scope

The computation records the two exact slopes, their gap and ratio, the tail exponent at a_{loc} , and the clearance exponents at a_{tail} and $a = 4$. No endpoint traces are fitted.

Remark 3.1. The theorem blocks only the present isolated-interior-cycle proof. It does not prove that the physical noisy trace fails at logarithmic order, and it does not exclude cancellations, boundary-layer localization, or a direct contour aggregate. Gates A–E remain false/open.